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Review

The effects of deforestation on the hydrological cycle in Amazonia: a review on scale and resolution

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Abstract:

This paper reviews the effects of deforestation on the hydrological cycle in Amazonia according to recent modeling and observational studies performed within different spatial scales and resolutions. The predictions that follow from future scenarios of a complete deforestation in the region point to a restrained water cycle, while the simulated effects of small, disturbed areas show a contrasting tendency. Differences between coarsely spatially averaged observations and finely sampled data sets have also been encountered. These contrasts are only partially explained by the different spatial resolutions among models and observations, since they seem to be further associated with the weakening of precipitation recycling under scenarios of extensive deforestation and with the potential intensification of convection over areas of land-surface heterogeneity. Therefore, intrinsic and interrelated scale and heterogeneity dependencies on the impact of deforestation in Amazonia on the hydrological cycle are revealed and the acknowledgement of the relevance of these dependencies sets a few challenges for the future. Copyright © 2007 Royal Meteorological Society

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INTRODUCTION

Land-surface changes may affect climate and, consequently, the hydrological cycle (Charney *et al.*, 1975; Eagleson, 1978; Eagleson, 1982; Williams and Balling, 1996). Water flux anomalies linked to these changes have already been detected in many parts of the globe, such as Yangtze (Yin and Li, 2001; Yang *et al.*, 2002), Mekong (Goteti and Lettenmaier, 2001) and Mississippi (Cherkauer *et al.*, 2000) river basins, as well as in several catchments in Africa (Calder *et al.*, 1995; Hetzel and Gerold, 1998; van Langenhove *et al.*, 1998). Recently, major land-surface changes have been particularly observed in the tropics (Aldhous, 1993), and Amazonia – which holds more than 40% of all remaining tropical rainforests in the world (Laurance *et al.*, 2001) – has been the focus of many studies about the impact of such changes on hydrological dynamics.

The Amazon basin (Figure 1) is the largest watershed in the world with a drainage area of ~7 million km^2 (Sioli, 1984a). Its strong and regular mainstem river is responsible for approximately 13% of the total global runoff into the oceans (Richey *et al.*, 1989b; Marengo *et al.*, 1994; Callede *et al.*, 2002; Dingman, 2002; Foley *et al.*, 2002). Its abundant vegetation releases large amounts of water vapor by transpiration, which, together with evaporation, equals 50–60% of the total rainfall in the region (Franken and Leopoldo, 1984; Vörösmarty *et al.*, 1989; Salati and Nobre, 1991; Victoria *et al.*, 1991). Part of this rainfall is sustained locally by evapotranspiration, induced by a precipitation recycling of about 25–35% (Brubacker *et al.*, 1993; Eltahir and Bras, 1994; Trenberth, 1999). The Amazonian rainforest thus

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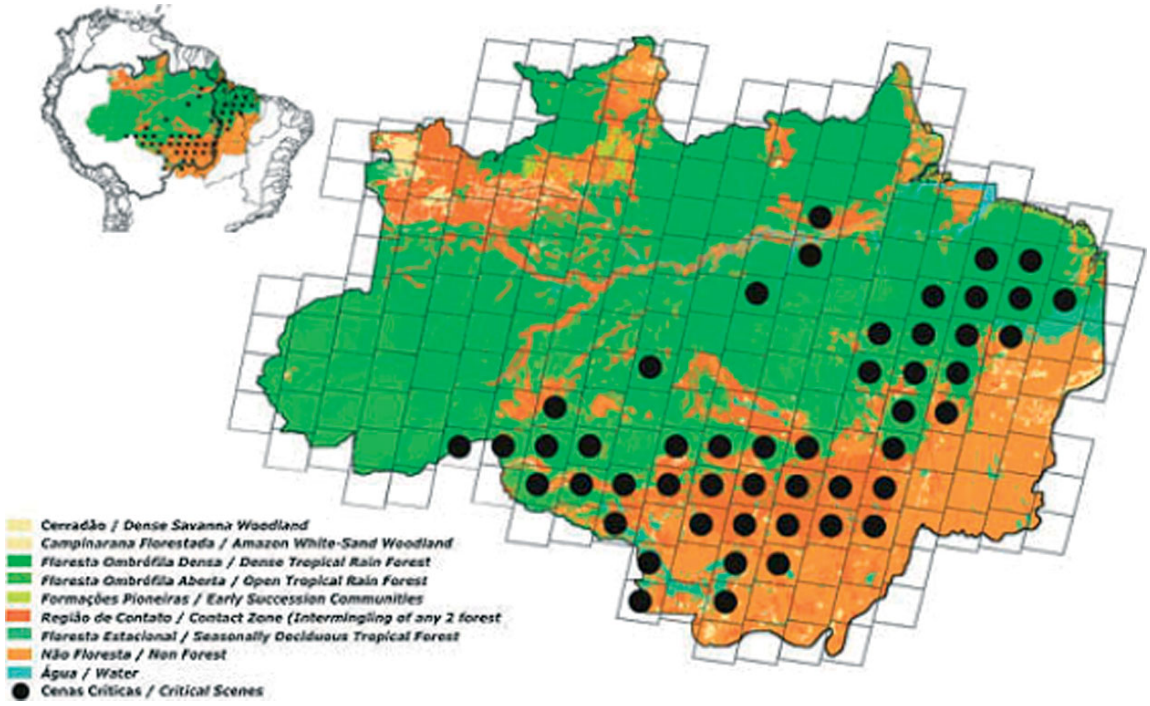

Figure 1. Vegetation types in Brazil's Legal Amazonia and spots (black dots) with the highest rate of deforestation in that area (main figure) (INPE, 2004), as measured by LANDSAT images. Geographic location (upper-left corner) of both Legal Amazonia and of the Amazon basin (thick lines) in northwestern South America. This figure is available in colour online at www.interscience.wiley.com/ijoc

considerably affects both water and energy balances in the basin, as well as both regional and global climates (Eagleson, 1978; Shukla *et al*., 1990; Nobre *et al*., 1991; Martinelli *et al*., 1996; Zeng *et al*., 1996; Werth and Avissar, 2002). Historically, land-surface changes in Amazonia intensified in the mid and early 1970s, when strategic governmental plans (e.g. Brazil's 'Programa de Integração Nacional') first attempted to promote the economic development across the region. These plans included the construction of extensive roads throughout the basin and the implementation of fiscal incentives for new settlers, triggering a massive migration of landless people into the region (Kelly and London, 1983; Moran, 1993). Since then, deforestation has become an intensive activity within the basin (Millet *et al*., 1998; Peterson and Heemskerk, 2001; Steininger *et al*., 2001), and, by the early 1990s, more than 10% of the basin's original forest had been converted to pasture or cropland (Fearnside, 1993), and, more recently, preferably to soybean culture (Fearnside, 2001). In Brazilian Amazonia alone, deforestation has reached an average rate of 1.78×10^4 km^2/year from 1988 to 2003 (INPE, 2004). However, despite all the concern and awareness of the scientific community with deforestation in Amazonia – evidenced through projects ABRACOS (Gash *et al*., 1996) and (LBA, 1996; Silva Dias *et al*., 2002), among others, – there is still some disagreement among predictions and observations regarding its effects on the water cycle in the region. This is especially due to the wide range of approaches employed, associated with different spatial scales and resolutions.

Many macroscale modeling studies have simulated a complete deforestation in Amazonia, typically predicting reductions in precipitation, evapotranspiration, moisture convergence and (possibly) runoff, along with increments in surface temperature. However, this outcome is not strictly consistent with findings from various mesoscale model studies, which have continually suggested an increase in convection and potential rainfall along the borders between forested and deforested areas. In a similar manner, apparently conflicting results have also been encountered by observational studies pursued at different scales. Enhanced overland flow has been observed over disturbed catchments in Amazonia, while significant trends on river discharge records collected close to the mouth of the basin have not been reliably observed yet. Identification of these contrasts prompts us to challenge either the adequacy of the numerical models employed or the accuracy of the observations performed – or even both. However, there are factors not related to the consistency of either models or observations that may satisfactorily explain such contrasts. On the basis of the size of Amazonia and on the importance of its vegetation to climate, the overall hydrological impact of deforestation seems to depend on both extent and spatial heterogeneity of the disturbance, as a result of the distinct land–atmosphere interactions induced by each particular scenario. The present work thus gives an overview of major findings in the literature on this topic, focusing on the hypothesis of intrinsic and interrelated scale and spatial heterogeneity dependencies on the hydrological impact of deforestation, their causes and implications. At the end, all relevant aspects raised throughout the paper

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are summarized and a few related aspects requiring further attention by the scientific community are mentioned.

LESSONS FROM MODELING STUDIES

Several modeling studies have been conducted during the last few decades with the objective of understanding the impact of deforestation on the hydrological cycle in Amazonia. These studies have simulated different deforestation scenarios and measured their impact on several relevant variables. Depending on the spatial resolution of the model, and especially on the extent of the horizontal domain considered, these studies are cast at the macroscale ($>10^5$ km^2), or at the mesoscale (10^2–10^5 km^2). One-dimensional (vertical) models, known as single-column models (SCMs), are also used to simulate the atmospheric profile above disturbed and undisturbed sites.

Macroscale models

Numerous modeling studies have relied on atmospheric general circulation models (AGCMs) along with their land-surface schemes to simulate extreme scenarios of deforestation in Amazonia (Table I). Such scenarios are reproduced by adjusting appropriate parameters in the model accordingly, and the predictions encountered are then compared to those from an almost identical simulation, associated with no deforestation. The difference between predictions from both simulations at steady state then provides an estimation of the impact of deforestation, while the uncertainties generated by other factors are assumed to get mutually canceled.

The predictions encountered by such models indicate a long-term tendency for decreasing precipitation and evapotranspiration, and for increasing surface temperature. There is also an indication that runoff may decrease with deforestation, even though no definitive trend direction has been suggested. A conceptual model explaining the mechanism of large-scale deforestation was proposed by Eltahir (1996), who suggested that the reduction induced on the net surface radiation is the primary and dominating effect that triggers all subsequent changes on both water and energy cycles within the disturbed region, ultimately causing the weakening of the adjacent large-scale atmospheric circulation. The main factors involved in the decline of net radiation (von Randow *et al*., 2004) have been linked to reductions in surface roughness length and increments in albedo (Lean and Warrilow, 1989; Berbet and Costa, 2003). These variables are heavily dependent on the land-cover type (Culf *et al*., 1995; Federer *et al*., 1996) and thus change considerably with the replacement of mature forests by pastures, or croplands. Reductions in transpiration and canopy interception (Nepstad *et al*.,

Table I. Macroscale model simulations of extreme scenarios of deforestation in Amazonia and the predicted changes on mean surface temperature (T), total daily rainfall (P), evapotranspiration (E) and runoff (R). Numbers on the left refer to those in Figure 4(a).

Reference	AGCM	Resolution (lat × lon)	Simulation (months)	ΔP (mm/d)	ΔE (mm/d)	ΔR (mm/d)	ΔT (°C)
Lean and Warrilow, 1989	UKMO[a]	2.5° × 3.75°	36.0	−1.43	−0.85	−0.40	+2.40
Nobre *et al*., 1991	NMC[b]	2.5° × 3.75°	12.5	−1.76	−1.36	−0.40	+2.50
Henderson-Sellers *et al*., 1993	CCM1[c]	4.5° × 7.5°	72.0	−1.61	−0.64	−0.90	+0.60
Lean and Rowntree, 1993	UKMO[a]	2.5° × 3.75°	36.0	−0.81	−0.55	−0.20	+2.10
Dirmeyer and Shukla, 1994	COLA[d]	4.5° × 7.5°	48.0	+0.24	−0.31	+0.02	+2.00
Polcher and Laval, 1994a	LMD[e]	2.0° × 5.6°	13.5	+1.08	−2.07	+3.70	+3.80
Polcher and Laval, 1994b	LMD[e]	2.0° × 5.6°	132.0	−0.51	−0.35	−0.16	+0.14
Sud *et al*., 1996	GLA	4.0° × 5.0°	36.0	−1.48	−1.22	−0.26	+2.00
Manzi and Planton, 1996	EMERAUDE[f]	2.8° × 2.8°	50.5	−0.40	−0.31	+0.33	−0.50
Lean *et al*., 1996	HC[g]	2.5° × 3.75°	120.0	−0.43	−0.81	+0.39	+2.30
Lean and Rowntree, 1997	HC[g]	2.5° × 3.75°	120.0	−0.27	−0.76	+0.51	+2.30
Hahmann and Dickinson, 1997	CCM2[h]	2.8° × 2.8°	120.0	−0.99	−0.41	−0.50	+1.00
Costa and Foley, 2000	GENESIS[i]	4.5° × 7.5°	180.0	−0.70	−0.60	−0.10	+1.40
Kleidon and Heimann, 2000	ECHAM4[j]	5.6° × 5.6°	240.0	−0.38	−1.30	+0.92	+2.50
Voldoire and Royer, 2004	ARPEGE[k]	2.8° × 2.8°	360.0	−0.40	−0.40	−0.01	−0.01

[a] *United Kingdom Meteorological Office*; Slingo *et al*. (1989).
[b] *National Meteorological Center*; Sela (1980); Kinter *et al*. (1988).
[c] Community Climate Model v.1; Williamson *et al*. (1987); Williamson and Williamson (1987).
[d] *Center for the Ocean-Land-Atmosphere Studies*; Sela (1980); Kinter *et al*. (1988).
[e] *Laboratoire de Météorologie Dynamique*; Sadourny and Laval (1984); Laval and Picon (1986).
[f] *Météo-France* spectral model; Ernie (1985); Coiffier *et al*. (1987); Geleyn *et al*. (1988).
[g] *Hadley Center*; Jones *et al*. (1995).
[h] Community Climate Model v.2; Hack *et al*. (1993).
[i] Pollard and Thompson (1995); Thompson and Pollard (1995a,b).
[j] Roeckner *et al*., 1996.
[k] Déqué (1999).

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1994; Hodnett *et al.*, 1995) are also linked to deforestation, all leading to a decrease in evapotranspiration, and, especially in Amazonia, further contributing to a decline in rainfall due to the strong precipitation recycling in the region (Franken and Leopoldo, 1984; Salati and Nobre, 1991). The projected reduction in runoff follows directly from the magnitude of the predicted change in precipitation (Lean and Rowntree, 1993), which, at least at the basin scale, is expected to be greater than the predicted change in evapotranspiration (Nobre *et al.*, 1991). Furthermore, the fact that the predicted runoff is equivalent to the difference between two large quantities such as precipitation and evapotranspiration – therefore carrying the uncertainty from both predicted values – may help explain the range of predicted values for this variable as shown in Table I. More importantly, runoff at the mouth of the basin must be equal to the water vapor convergence at steady state in long-term model runs, and, since this term is normally considered a boundary condition for the integrations, it naturally induces distinct tendencies to runoff – and to some extent, to precipitation and evapotranspiration as well – for each large-scale atmospheric circulation scenario employed.

The prediction of enhanced surface temperature is consistent with the increase in Bowen ratio – which equals the ratio of sensible to latent heat flux – observed after deforestation (Nobre *et al.*, 1991). The daily variability of surface temperature in Amazonia is also expected to increase following a complete deforestation in the region, even when its long-term mean does not change significantly (Voldoire and Royer, 2004). Other relevant changes associated are reductions in plant-available water capacity (Zhang *et al.*, 2001) and in infiltration capacity (Bruijnzeel, 1996), respectively, due to reduced root-zone depth over pastures (Nepstad *et al.*, 1994) and to soil compaction during and after clearing. In fact, the decline in rooting depth induced by deforestation has even been suggested to be the main factor affecting the climate in Amazonia (Kleidon and Heimann, 2000). An increase in the stomatal resistance is another anticipated result of deforestation, which, together with all other concomitant predictions, may contribute to the lengthening of the dry season in the Amazon Region (Shukla *et al.*, 1990), which is the period when the effects of deforestation are more severe (Silva Dias *et al.*, 2002). Following this and several other anticipated positive feedbacks, it has been suggested that a complete and rapid destruction of the tropical forests in Amazonia could lead to irreversible climatic changes in the region (Nobre *et al.*, 1991; Oyama and Nobre, 2003). Significant climatic changes are further expected in remote parts of the globe through the establishment of teleconnection patterns induced by the atmospheric disturbances generated by a complete deforestation in Amazonia (Salati and Nobre, 1991; Werth and Avissar, 2002). Additionally, changes on cloud coverage and surface albedo induced by biomass fire emissions (Fisch *et al.*, 1994) and the climate-driven forest dieback associated with scenarios of global warming (Cox *et al.*, 2004) are expected to affect both energy and water balances inside the basin (Dickinson and Kennedy, 1992; Betts *et al.*, 2004; Huntingford *et al.*, 2004).

Mesoscale models

Simulation of the effects of deforestation by mesoscale models enables the assessment of finer-scale land–atmosphere feedbacks that are not accurately resolved by models with much coarser spatial resolutions. Atmospheric instabilities induced between areas of forest and pasture (Dolman *et al.*, 1999; Liu *et al.*, 1999; Baidya Roy and Avissar, 2000; Souza *et al.*, 2000; Weaver and Avissar, 2001) are thus better represented by mesoscale models, which have showed that the impact of such instabilities are (typically) quite different from the results encountered by AGCM simulations of a basin-wide deforestation (Table II).

Various observational studies (reviewed by Segal *et al.*, 1988) detected mesoscale anomalous circulations induced by air-temperature contrasts over regions of extreme land-surface gradients in different parts of the globe. In Amazonia, such circulations are expected to be observed especially during the dry season, when contrasts in soil moisture conditions and therefore on the convective boundary layer (CBL) depth over forests and pastures are greater (Fisch *et al.*, 2004). Modeling studies have tried to reproduce that effect and it has been noted that such circulations may significantly affect the timing and formation of clouds, potentially altering both intensity and distribution of precipitation (Chen and Avissar, 1994). It has been estimated that, at the mesoscale, a landscape with a relatively large discontinuity tends to produce more precipitation than a homogeneous domain, inducing a negative feedback that ultimately tends to eliminate the discontinuity (Avissar and Liu, 1996). In some cases, the thermal circulation induced may become as intense as a sea-breeze circulation, for example, over domains with extended areas of unstressed dense vegetation bordering areas of bare soil (Segal *et al.*, 1988). The horizontal scale of such landscape heterogeneities is another factor that may affect the establishment of precipitation (Pielke *et al.*, 1991), while the optimum scale for triggering convection seems to depend on the air-humidity level (Avissar and Schmidt, 1998). A strong enough synoptic (or background) wind-field may also interact with the induced circulation, possibly masking its existence at times (Segal *et al.*, 1988). It was noted that a mild background wind of 5 ms^{-1} may be sufficient to virtually remove all thermal impacts generated by the land-surface discontinuities (Avissar and Schmidt, 1998), although more recent studies have revealed that a strong background wind may only advect the instabilities elsewhere rather than disperse them completely (Baidya Roy and Avissar, 2002). The detection of such aspects at the mesoscale leads to a contrast to the predictions of macroscale models that had been suggested by Eltahir and Bras (1994) earlier, who simulated a single deforested area of moderate size ($\sim 6 \times 10^4$ km^2) in west-central Amazonia with a mesoscale model and predicted

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Table II. Mesoscale model simulations of atmospheric conditions above deforested areas within Amazonia. Numbers on the left refer to those in Figure 4(a).

Reference	Mesoscale model	Resolution (km × km)	Simulation (days)	Grid center	Key findings
Eltahir and Bras, 1994	MM4[a]	50 × 50	93	6.5 °S, 67.5 °W	Less rainfall, less evaporation
Silva Dias and Regnier, 1996	RAMS[b]	20 × 20	4	~10 °S, 60 °W	Greater vertical motion
Dolman *et al*., 1999	RAMS[b]	16 (4, 1) × 16 (4, 1)[e], 60 (20) × 60 (20)[d]	4	~10.5 °S, 62 °W	Deeper convective layer
Wang *et al*., 2000	MM5V2[c]	12 (4) × 12 (4)[e]	6	~11 °S, 63 °W	More convection during dry-season
Baidya Roy and Avissar, 2002	RAMS[b]	16 (4, 1) × 16 (4, 1)[e]	1	10 °S, 62.5 °W	More convection triggered by surface heterogeneity
Tanajura *et al*., 2002	ETA/SSiB[d]	80 × 80	30	22 °S, 60 °W	Less rainfall, less evaporation
Weaver *et al*., 2002	ClimRAMS[b]	16 (4, 1) × 16 (4, 1)[e], 16 (4, 2) × 16 (4, 2)[e], 16 (4, 4) × 16 (4, 4)[e]	2	~10 °S, 62 °W	Effects predicted depend on correct model configuration

[a] Giorgi (1990).
[b] Pielke *et al*. (1992).
[c] Grell *et al*. (1994).
[d] Xue *et al*. (1996).
[e] Nested grids.

a weaker decline on the water cycle in comparison with most macroscale modeling studies. Correspondingly, an ensemble of extensive scenarios of deforestation performed with a mesoscale model has predicted a stronger impact in comparison to most macroscale simulations of similar scenarios (Tanajura *et al*., 2002).

The application of mesoscale models to portions of Amazonia have enabled the evaluation of the effects of land-surface discontinuities under an actual scenario of deforestation. Extensive areas of native forests within the state of Rondônia (in the southwestern part of Brazilian Amazonia) have been extensively replaced by pastures (Skole and Tucker, 1993), making it one of the sites of application of such gridded models. Especially in the dry season, it has been noted that the interaction between mesoscale circulations induced by land-surface heterogeneities and the large-scale flow may enhance and deepen the convective activity over disturbed areas (Baidya Roy and Avissar, 2002), in agreement with cloud cover surveys performed by Cutrim *et al*. (1995). During the rainy season, however, deforestation in Rondônia does not seem to have a significant effect on the distribution of cloudiness and rainfall, since the synoptic conditions tend to be propitious to induce mesoscale circulations alone (Wang *et al*., 2000), in agreement with the satellite images evaluated by Laurent *et al*. (2002). The influence of topography (Silva Dias and Regnier, 1996), coastlines and large rivers within Amazonia in the formation of mesoscale circulations should also be taken into consideration, possibly through the application of nested models (Gandu *et al*., 2004).

Single-column Models (SCMs)

The use of SCMs at a few points in Amazonia (Table III) has enabled the investigation of the vertical structure of the atmosphere above both disturbed and undisturbed sites. This approach has helped in clarifying the impact of these scenarios on the local convective activity, even though this type of model neglects the horizontal interactions caused by the surrounding land-surface discontinuities.

As a result of the higher evapotranspiration flux released by undisturbed areas, Rocha *et al*. (1996) encountered greater convective precipitation over forested areas in Amazonia than over pastures. In a similar assessment, however, Fisch *et al*. (1996) simulated a deeper CBL over pasture, compared to nearby forest sites in Rondônia. Still, both timing and depth of the CBL seems to have been significantly underestimated over pasture, when compared with observations made concurrently at the same sites (Nobre *et al*., 1996), arguably due to the inability of one-dimensional models to reproduce the thermal instabilities induced across the surrounding deforested strips. Similar results were encountered by Dolman *et al*. (1999), who noted that modeling CBL over pastures in Rondônia may not only make it seem lower than observations (Calvet *et al*., 1997) but also colder and wetter, indicating the failure of SCMs to generate the necessary amount of heat to induce a deeper and warmer CBL. These findings were supported by additional experiments performed by Dolman *et al*. (1999), who showed that even mesoscale gridded models may fail to properly predict both depth and temperature of the CBL over pastures in Rondônia, despite their ability to simulate the

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Table III. Single-column model (SCM) simulations of atmospheric conditions above forested and deforested sites in Brazilian Amazonia.

Reference	SCM	Study sites	Simulation (*h*)	Period of simulations	Key findings
da Rocha *et al*., 1996	SiB-1D[a]	2°19′S, 60°19′W[b]; 2°57′S, 59°57′W[c]; 10°45′S, 62°22′W[d]	52	July 1993	More convection over forest
Fisch *et al*., 1996	CBL type[e]	10°05′S, 61°55′W[f]; 10°45′S, 62°22′W[c]	9	July 1993	Deeper CBL over pasture
Dolman *et al*., 1999	MESO-NH[g]	10°05′S, 61°55′W[f]; 10°45′S, 62°22′W[c]	12	August 1994	Deeper CBL over pasture

[a] Sellers *et al*. (1986).
[b] Fazenda Dimona, Amazonas; pasture (surrounded by forest).
[c] Reserva Ducke, Amazonas; forest.
[d] Fazenda Nossa Senhora Aparecida, Rondônia; pasture.
[e] Tennekes (1973).
[f] Reserva Jaru, Rondônia; forest (adjacent to pasture).
[g] Lafore *et al*. (1998).

anomalous convection and sensible heat fluxes caused by surrounding land-surface heterogeneities.

LESSONS FROM OBSERVATIONAL STUDIES

The hydrological impact of deforestation in Amazonia has also been evaluated through observational studies, aimed at detecting significant changes on the water cycle in the basin that may be linked to the effects of clearing. These studies have focused on either small ($<10^2$ km^2)–at catchments and point sites – or large areas ($>10^2$ km^2) – at basin and subbasin scales.

Basin and subbasin scale observations

Several studies have searched for significant trends in the mean hydrological cycle in Amazonia through the application of a variety of trend analysis methods to a diverse set of time series recorded over the last century (Table IV). The collection of results obtained denied the existence of mean trends in the basin, since they have not been consistently detected with significance. Furthermore, such observations have not agreed with the general predictions from macroscale simulations of deforestation.

Increasing trends in discharge and precipitation were observed at all but the eastern parts of the Amazon basin between the late 1950s and the early 1980s (Rocha *et al*., 1989). However, despite contentions that these trends were associated with upstream areas of deforestation (Gentry and Lopez-Parodi, 1980), most time series retreated to their long-time means by the end of the period (Marengo, 1995). In support of previous criticisms (Nordin and Meade, 1982), it has been suggested that the variability observed in both Amazonian rainfall and discharge time series during that period was a response to fluctuations over the Tropical Pacific, associated with El Niño Southern Oscillation (ENSO) events (Richey *et al*., 1989a; Marengo *et al*., 2001) and not deforestation. In fact, apart from the remote effect of the interannual anomalies of SST from both Atlantic and Pacific Oceans (Marengo *et al*., 1993; Marengo *et al*., 1998), the interdecadal climate variability in Amazonia may be further influenced by the global divergent circulation, which appears to be intensifying the water cycle in Amazonia since (at least) the late 1950s (Chen *et al*., 2001). Additionally, Chu *et al*. (1994) have detected significant trends of decreasing outgoing long-wave radiation (OLR) (associated with enhanced convection) in the western part of the basin between the mid 1970s and the early 1990s, together with nearly significant increasing rainfall trends at both central and eastern parts of the basin. More recently, Marengo (2004) tested for trends on long-term rainfall data in Amazonia and the only significant signal encountered refers to weak decreasing trends, especially in the northern part of the basin, where virtually no clearing activities have been performed yet. These findings thus support the idea that the atmospheric fluctuations induced by remote forcings (Richey *et al*., 1989a; Fu *et al*., 2001) can potentially offset or overshadow the effects of deforestation (Chen *et al*., 2001).

The existence of trends on additional terms of the hydrological cycle in Amazonia have also been tested, and significant changes on spatial averages for the input and output fluxes of water vapor (decreasing) and for precipitation recycling (increasing) were encountered (Costa and Foley, 1999). However, as suggested by Paiva and Clarke (1995), the use of spatially aggregated point data may not be appropriate for the detection of trends, owing to the inevitable 'dilution' of the signal during the upscaling process. In fact, despite the significant changes encountered on mean discharge in the Tocantins basin, a sizable watershed ($>10^5$ km^2) on eastern Amazonia, the comparison between hydrological records from periods of low (1949–68) and high (1979–98) land-surface disturbances have not shown significant changes on spatially averaged precipitation (Costa *et al*., 2003). The

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Table IV. Observations aimed at detecting trends in the hydrological cycle in Amazonia at basin and subbasin scales. Numbers on the left refer to those in Figure 4(b).

Reference	Domain of interest	Data sets	Time span	Key findings
da Rocha *et al*., 1989	Amazon basin	24 ANA[b], CNEC[c], SENAMHI[d] stations (2p, 22f)[a]	1903–1986	No consistent trend
Richey *et al*., 1989a	Negro, Solimões subbasins	1 ANA[b], 1 PORTOBRAS[e] stations (1w, 1f)[a]	1903–1985	No consistent trend
Chu *et al*., 1994	Amazon basin	2 stations[f] (p)[a]; NOAA[g] OLR (g)[a]	1974–1990	Increase in convection
Paiva and Clarke, 1995	Amazon basin	48 ANA[b] stations (48p)[a]	1960s–1990s	No consistent trend
Marengo, 1995	Negro subbasin	1 ANA[b] station (1w)[a]	1903–1992	No consistent trend
Marengo *et al*., 1998	Amazon basin	16 ELETROBRAS[h], ELETRONORTE[i] stations (8p, 8f)[a]	1930s–1990s	No consistent trend
Costa and Foley, 1999	Amazon basin	NCEP/NCAR[i] (g)[a]	1976–1996	Increase in recycling
Chen *et al*., 2001	Amazon basin	GHCN[l] stations (p, t, pr)[a]; SST[l], NCEP/NCAR[j], NOAA[g] OLR (g)[a]	1950s–1990s	Increase in rainfall
Costa *et al*., 2003	Tocantins basin	1 ANA[b] station (1f)[a]; CRU[m] (g)[a]	1949–1998	Increase in discharge
Durieux *et al*., 2003	Arc of deforestation	ISCCP[n], GPCP[o], TRFIC[p] (g)[a]	1984–1993	Increase in seasonality
Marengo, 2004	Amazon basin	~300 GHCN[k], INMET[q], CPTEC[r], ANA[b] stations (p)[a]; CRU[m], CMAP[s] (g)[a]	1929–1998	Decrease in rainfall
Negri *et al*., 2004	Southwestern Amazonia	GHCN[l] stations (p)[a]; GOES[t] TMI[u], SSM/I[v] (g)[a]	1960–1990	Increase in rainfall

[a] p = pluviometric, f = fluviometric, t = temperature, pr = pressure, w = water level, g = gridded data.
[b] Agência Nacional de Águas.
[c] Consórcio Nacional de Engenheiros Consultores S.A.
[d] Servicio Nacional de Meteorologia e Hidrologica.
[e] Empresa de Portos do Brasil S.A.
[f] Chu (1991).
[g] National Oceanic and Atmospheric Administration.
[h] Centrais Elétricas Brasileiras S.A.
[i] Centrais Elétricas do Norte do Brasil.
[j] National Centers for Environmental Prediction/National Center for Atmospheric Research reanalysis data.
[k] Global Historical Climatology Network.
[l] Smith and Reynolds (1998).
[m] Climate Research Unit.
[n] International Satellite Cloud Climatology Project; Rossow and Schiffer (1991).
[o] Global Precipitation Climatology Project; Huffman *et al*. (1997).
[p] Tropical Forest Information Center; TRFIC (2000).
[q] Instituto Nacional de Meteorologia.
[r] Centro de Previsão de Tempo e Estudos Climáticos.
[s] CPC Merged Analysis of Precipitation.
[t] Geostationary Operational Environmental Satellite.
[u] Tropical Rainfall Measuring Mission Microwave Imager.
[v] Special Sensor Microwave Imager.

precipitation record used in this study refers to a rather coarsely ($0.5^\circ \times 0.5^\circ$) gridded data set (New *et al*., 2000) and, therefore, it is unclear whether significant changes on precipitation would still be absent in case they had been monitored on a finer-scale. Similarly, rainfall estimates made along the Amazon arc of deforestation using a $2.5^\circ \times 2.5^\circ$ grid did not seem to be influenced by deforestation (Durieux *et al*., 2003), while concurrent estimates gathered using a $0.5^\circ \times 0.5^\circ$ grid suggested an increase in precipitation in northern Rondônia (Negri *et al*., 2004). Thus, taking into account current data resolution, abundance and quality, one cannot be entirely sure whether deforestation is affecting the water cycle in Amazonia, since the inherent effects could be occurring at subgrid, undetectable scales (Marengo, 1995).

Catchment and point observations

Field experiments have measured key hydraulic properties and water flux rates on both disturbed and undisturbed sites in Amazonia while trying to estimate the hydrological effects of clearing activities at small scales within the basin (Table V). The observations are in reasonable agreement with general expectations of enhanced water yield over cleared sites (Bosch and Hewlett, 1982; Oyebande, 1988; Sahin and Hall, 1996; Tucci and Clarke, 1997), a pattern that follows directly from the observed

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Table V. Catchment[k] and field studies on the hydrological impact of deforestation on different types of land cover in Brazilian Amazonia. Numbers on the left refer to those in Figure 4(b).

Reference	Study sites	Period of records	Sites land-cover	Key findings
(13[b], 14[c]) Franken and Leopoldo, 1984[k]	Central Amazonia[a]	1976–77, 1981–82[b], 1980–81[c]	Forest	More runoff, less rainfall
Shuttleworth, 1988a	Central Amazonia[a]	September 1983 – September 1985	Forest	Less evapotranspiration
Nepstad *et al*., 1994	Northeastern Amazonia[d]	June 1992 – October 1992	Forest, adjacent pasture	Less evapotranspiration
Hodnett *et al*., 1995	Central Amazonia[e,a]	1990–91[e]; 1970–93[a]	Pasture, adjacent forest	Less water uptake, more surface runoff
(15[g], 16[h]) Williams and Melack, 1997[k]	Central Amazonia[f]	July 1989 – July 1990	Forest, partially deforested	More runoff, less evapotranspiration
Hölscher *et al*., 1997	Northeastern Amazonia[i]	April 1992 – April 1993	Secondary forest	Fast recover on evapotranspiration
Jipp *et al*., 1998	Northeastern Amazonia[d]	1991–1994	Forest, secondary forest, pasture	More runoff, less evapotranspiration
Elsenbeer *et al*., 1999	Southwestern Amazonia[j]	1984–1995	Forest, pasture, plantation	More surface runoff

[a] Reserva Ducke, 25 km north of Manaus, Amazonas.
[b] Barro Branco Watershed.
[c] Bacia Modelo Watershed.
[d] Fazenda Vitória, Paragominas, Pará.
[e] Fazenda Dimona, 100 km north of Manaus, Amazonas.
[f] Lake Calado, 80 km west of Manaus, Amazonas.
[g] Braço do Mota Watershed.
[h] Igarapé de Mota Watershed.
[i] Igarapé Açu, Pará.
[j] Rancho Grande, Rondônia.

reduction in evapotranspiration – arising predominantly from declines in transpiration, interception and water uptake.

Following the observation of large amounts of interception and transpiration over selected undisturbed catchments in Amazonia, Franken and Leopoldo (1984) showed through water budget calculations that deforestation in these areas would not only induce a decrease in evapotranspiration but also a huge increase in local runoff. On the basis of many field studies performed in the basin, Sioli (1984b) further noted that deforestation results in soil compaction, which then contributes to enhanced surface runoff due to the corresponding reduction in infiltration. In fact, it was observed that the intensity of rainfall during storm events normally exceeds the infiltration capacity in pastures, inducing both on-surface and below-surface runoff (Elsenbeer *et al*., 1999). Increased runoff and decreased evapotranspiration were also measured after the clearing of a small catchment in central Amazonia (Williams and Melack, 1997), in agreement with previous suggestions of a substantial decrease in evaporation following nearby land-cover disturbances (Shuttleworth, 1988a). Measures of soil water content in forest and pasture near Manaus further indicated a deeper and therefore more efficient water uptake by the forest, thus supporting higher evaporation rates in comparison with pastures, which, in turn, displayed a greater spatial variability of soil moisture due to redistribution of rainfall as surface runoff (Hodnett *et al*., 1995). Similar observations confirmed that, contrary to forests, pastures cannot sustain high indices of evapotranspiration during extended periods of drought (Wright *et al*., 1992; Jipp *et al*., 1998; von Randow *et al*., 2004). Reductions in leaf canopy and root zone depth following deforestation have also been observed to diminish evapotranspiration and increase runoff (Nepstad *et al*., 1994; Tobón Marin *et al*., 2000). Therefore, unlike the general pattern at the basin scale, the water fluxes within small deforested sites seem to depend on local land-surface characteristics rather than on remote forcings – in agreement with the idea that, at small scales, the natural variability induced locally overcomes the magnitude of globally induced signals (Trenberth, 1997).

Secondary (regenerating) forests account for about 30% of the accumulated deforested area in Amazonia (Skole *et al*., 2002), and a few other field experiments have been conducted over such sites. Measurements taken over a 2.5-year-old secondary forest in the eastern part of the basin showed intermediate values of evaporation compared to typical estimates for pastures and primary

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forests (Hölscher *et al*., 1997). More recently, it has been observed that a nearby site with slightly more mature vegetation (3.5 years old) may be able to release evapotranspiration rates similar to those of forests (Sommer *et al*., 2002). Furthermore, measures of the saturated hydraulic conductivity under different land-surface areas – forest, secondary forest and pasture – showed that the hydraulic properties of the corresponding soil profiles are similar below all three sites – at least, between 20 and 50 cm depths (Godsey and Elsenbeer, 2002). Therefore, the shifting patterns of clearing and regrowth are likely to complicate efforts at examining land-use induced hydrology changes.

SUMMARY AND DISCUSSION

On the basis of the predictions of many AGCM studies, the expectation of a less intense water cycle in Amazonia following a basin-wide scenario of deforestation emerges. However, this expectation has not been confirmed by simulations of moderately sized scenarios of clearing, as many mesoscale modeling studies have shown. As to the observations performed in the region, none of the basin scale studies reviewed has encountered broad or significant changes on the hydrological cycle in Amazonia that could be directly and consistently associated with the effects of deforestation. At the same time, it has been reported that, at the catchment scale, the removal of the forest cover leads to enhanced runoff and decreased evapotranspiration.

On the basis of these findings, it has been proposed that deforestation in Amazonia seems to induce contrasting effects, depending on the spatial scale associated with the observed or simulated disturbance (D'Almeida *et al*., 2006). The primary cause for such a dependency is not strictly conceptual, but also operational. It relates to the fact that coarse resolution models cannot resolve small-scale phenomena with the same degree of detail as more refined models do. The same principle applies to observations, which may represent any particular phenomenon differently, depending on the grid resolution, or on the distribution of gauging stations available. Secondly, the considerable size of the Amazon basin together with the land–atmosphere interactions occurring within, cause opposing factors to be dominant at different scales, and, therefore, a contrast naturally emerges. One of these factors is the intense precipitation recycling observed in the region, which makes the evapotranspiration flux released by the forests the main source of water to the local precipitation (Figure 2(a)). As a consequence, a drastic deforestation scenario would result in a severe restructuring of land–atmosphere dynamics (Figure 2(d)), partially explaining why most AGCMs have predicted weakened water fluxes as a result of extensive deforestation. Small and localized areas of clearing, however, are insufficiently large to induce such an impact (Figure 2(b)), even though the accumulation of the local changes caused by such small clearings is exactly what affects the precipitation recycling in the basin as deforestation expands. In fact, depending on the resolution at which the potential changes on precipitation are monitored, even larger areas of deforestation may seem uncoupled to climate (Costa *et al*., 2003). The second main factor linked to such scale dependency is the impact of land-surface spatial heterogeneities on the atmospheric circulation above mesoscale deforested areas. At this scale, strong gradients on the surface sensible heat flux may contribute to an increase in rainfall through the establishment of anomalous convective circulations (Figure 2(c)). In fact, the degree of heterogeneity is expected to be as important as the size of the disturbance to the formation of the anomalous circulations just mentioned (Pielke, 2001). Therefore, despite the fact that such anomalous circulations occur preferably around mesoscale areas of clearing, even a substantial disturbance at this scale may not be able to generate any of such anomalies above overly fragmented – or 'disorganized' (Shuttleworth, 1988b) – domains. Furthermore, according to Baidya Roy *et al*. (2003), although the land–atmosphere dynamics acts as a medium-band pass filter, enabling only anomalous circulations within a certain scale range to evolve, the degree of heterogeneity is still an important factor to determine whether these circulations develop at the first place. It then follows that

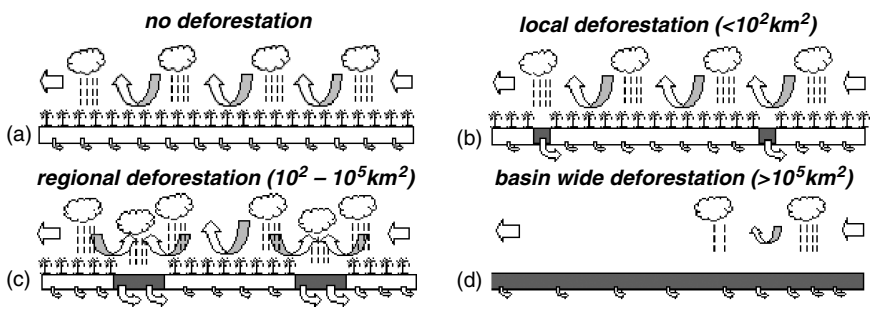

Figure 2. Schematic representation of the hydrological impact of different extents of clearing (in dark gray) in Amazonia. The horizontal water vapor flux transfers moisture into the region and in the case of (a) no deforestation, this flux is sustained by precipitation recycling, maintaining high indices of rainfall. Areas of (b) local deforestation are too small to affect rainfall, but runoff increases and evapotranspitation decreases. Areas of (c) regional deforestation are large enough to influence circulation, strengthening convection and potentially increasing rainfall. A (d) basin-wide deforestation scenario would impose a severe decline on evapotranspiration and then on precipitation recycling, weakening the hydrological cycle in Amazonia as a whole.

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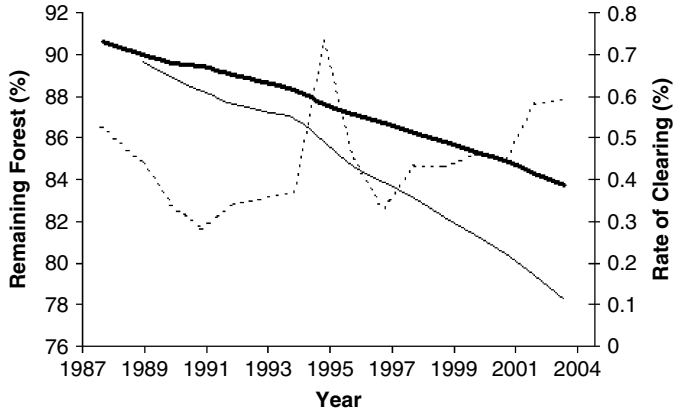

Figure 3. Percentage of remaining forest over an area of ~4 million km^2 in Brazil's Legal Amazonia in 1988–2003 (thick line), based on the mean annual rate of clearing (dashed line) estimated between consecutive LANDSAT scenes (INPE, 2004). The thin line indicates the percentage of forest that would have remained in the case of no regrowth within the basin, if the rate of clearing had been consistently equal to the net deforestation.

together with the aforementioned scale dependency on the impact of deforestation, there is also a heterogeneity dependency occurring, linked to the many different spatial distributions that a specific deforestation extent may display.

Directly from the acceptance of such dependencies, it follows that the downscaling of predictions from basin-wide scenarios of deforestation, or the upscaling of observations from disturbed catchment areas, may provide erroneous conclusions (Wood *et al*., 1988; Entekhabi *et al*., 1999). In addition, despite the high rates of cutting in the recent past, the size of the Amazon basin is still much larger than the extent of deforestation (Figure 3). Therefore, it is clearly premature for the predictions of extreme scenarios of deforestation to be effectively manifested or detected. Furthermore, extrapolating the predictions associated with an extreme and increasingly improbable (Baidya Roy and Avissar, 2002) scenario of complete deforestation to current conditions in Amazonia may not only interfere with investigations of actual trends in the basin but also negatively affect the policy-making process in the region. An unfruitful search for signs of a weakened water cycle may suggest that the ecosystems in Amazonia are not as sensitive to deforestation as they are to other important effects – like ENSO – which may dangerously contribute to the relaxation of government actions to slow down logging in Amazonia. Consequently, it seems that along with the simulation of such extreme scenarios, macroscale models should also acknowledge and represent the current distribution of deforestation and its effects (Gash *et al*., 2004), avoiding presently misleading expectations and enabling the check of predictions against observations. The correct simulation of water vapor convergence on long-term integrations due to its inevitable impact on runoff at steady state is also essential, requiring the evaluation of the sensitivity of the system to fluctuations on this term. In addition, the importance of correctly accounting for the extent and distribution of areas of recovering vegetation in Amazonia is addressed, since young secondary forests may be able to induce similar fluxes of water – depending on the plant species considered – in comparison with mature forests. Furthermore, an accurate representation of the extent of regrowth on abandoned pastures and ranches is crucial for a proper estimation of the net deforestation rate in the basin, since it is evident that the direct accumulation of the reported annual rates of clearing does not equal the actual decrease in forest coverage (Figure 3).

Moreover, many modeling studies tend to employ pure macroscale, or mesoscale approaches (Figure 4(a)), leaving gaps within the range of applicable spatial resolutions and simulation times. These gaps may be linked to the inability of conventional AGCMs to correctly reproduce relevant subgrid processes like the enhanced convection potentially induced over heterogeneously deforested areas in Amazonia. Such anomalous circulations are presently being generated on the mesoscale, but, since they may evolve to higher scales (Baidya Roy *et al*., 2003), they must in fact be adequately represented by AGCMs through their parameterization schemes (Bonell, 1998). However, despite the intense research on this topic (Avissar, 1992; Henderson-Sellers and Pitman, 1992; Koster and Suarez, 1992; Dickinson, 1996; Liu *et al*., 1999, among others), a consistent representation of these processes has not been widely adopted by the macroscale modeling community yet. The parameterizations employed by the current generation of AGCMs tend to rely only on the quantification of turbulence effects, neglecting the influence of the heat fluxes associated with anomalous mesoscale circulations (Baidya Roy and Avissar, 2002).

Regarding the absence of significant and consistent signs of deforestation in Amazonia among the studies reviewed, the recent decline of the world's gauging station network (IAHS, 2001) – a condition especially evident in remote areas such as Amazonia (ANA, 2001) – is certainly an issue. In fact, virtually all observational studies performed in the region are restricted to wide, coarsely monitored sections of the basin, or to just a few, small catchment sites (Figure 4(b)). Evidently, the only way to

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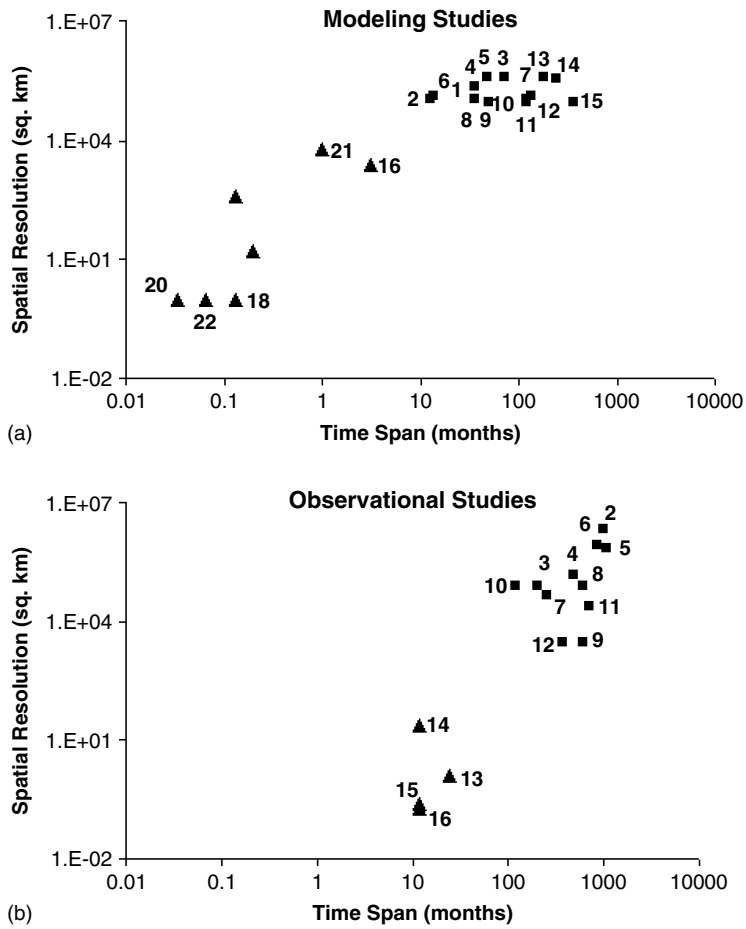

Figure 4. Distribution of the studies reviewed according to both spatial and temporal specifications of their (a) modeling experiments (squares refer to macroscale studies and triangles refer to mesoscale studies) and (b) observational approaches (squares refer to basin and subbasin scale studies and triangles refer to catchment and field studies). The numbers in the graphics refer to those shown in (a) Tables I and II and in (b) Tables IV and V.

overcome this situation is to develop a well-constituted gauging station network in Amazonia, which may be achieved by governmental initiatives such as Brazil's SIVAM project, ideally capable of detecting the contrast between localized and spatially aggregated effects of deforestation. However, due to the characteristics of the river network and to the asymmetric expansion of deforestation in Amazonia, there are portions of the basin that are more susceptible to the potential effects of deforestation than others (Sombroek, 2001; Fearnside, 2005). The identification of such strategic areas would then increase the effectiveness of such improvements in the network by strengthening potential sings of deforestation, in spite of the superimposed signal induced by remote forcings.

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